

Chapter 9

Image and Video segmentation

IMP302m

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Fundamentals of Segmentation

- The fundamental problem in segmentation is to partition an image into regions that satisfy the preceding conditions.
- Segmentation algorithms for monochrome images generally are based on one of two basic categories dealing with properties of intensity values: *discontinuity* and *similarity*
- *Discontinuity*: boundaries of regions are sufficiently different from each other, and from the background, to allow boundary detection based on local discontinuities in intensity: Edge-based segmentation
- *Similarity*: partitioning an image into regions that are similar according to a set of predefined criteria: Region-based segmentation

Fundamentals of Segmentation

- Remind from CPV301
- Point, line and edge detection
- K-means, Gaussian mixture model, Mean shift
- Watershed, Split and Merge, Graph-based, Probabilistic aggregation

9.1 Clustering and superpixels based segmentation

- **Region segmentation using K -means clustering**

- Let $\{z_1, z_2, \dots, z_Q\}$ be set of vector observations (samples). These vectors have the form $\mathbf{z} = [z_1 \ z_2 \ \dots \ z_n]^T$

- grayscale image: z is scalar; color image: z is 3-D vector

- Partition the set Q of observations into k ($k \leq Q$) disjoint cluster sets

$$C = \{C_1, C_2, \dots, C_k\}$$

$$\arg \min_C \left(\sum_{i=1}^k \sum_{\mathbf{z} \in C_i} \|\mathbf{z} - \mathbf{m}_i\|^2 \right)$$

where $\mathbf{m}_i$ is the mean vector of the samples in set C_i

9.1 Clustering and superpixels based segmentation

- **Region segmentation using K -means clustering**

1. ***Initialize the algorithm:*** Specify an initial set of means $\mathbf{m}_i(1)$, $i = 1, 2, \dots, k$
2. ***Assign samples to clusters:*** Assign each sample to the cluster set whose mean is the closest (ties are resolved arbitrarily, but samples are assigned to only *one* cluster):
$$\mathbf{z}_q \rightarrow C_i \text{ if } \|\mathbf{z}_q - \mathbf{m}_i\|^2 < \|\mathbf{z}_q - \mathbf{m}_j\|^2 \quad j = 1, 2, \dots, k (j \neq i); q = 1, 2, \dots, Q$$
3. ***Update the cluster centers (means):***
$$\mathbf{m}_i = \frac{1}{|C_i|} \sum_{\mathbf{z} \in C_i} \mathbf{z} \quad i = 1, 2, \dots, k$$
4. ***Test for completion:*** Compute the Euclidean norms of the differences between the mean vectors in the current and previous steps. Compute the residual error, E , as the sum of the k norms. Stop if $E \leq T$, where T a specified, nonnegative threshold. Else, go back to Step 2

9.1 Clustering and superpixels based segmentation

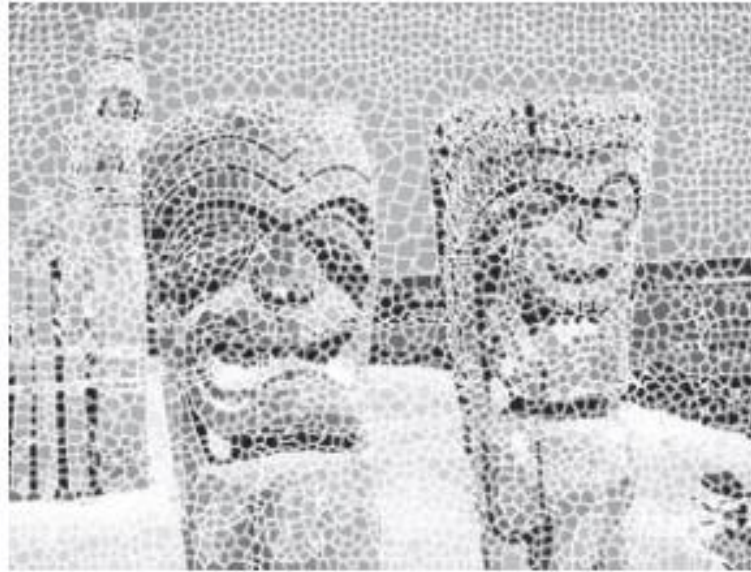
- **Region segmentation using superpixels**
- Replace the standard pixel grid by grouping pixels into primitive regions that are more perceptually meaningful than individual pixels
- Objectives: lessen computational load, and to improve the performance of segmentation algorithms by reducing irrelevant detail
- One important requirement of any superpixel representation is *adherence to boundaries* → boundaries between regions of interest must be preserved in a superpixel image

9.1 Clustering and superpixels based segmentation

- Region segmentation using superpixels



(a) Image of size 600×480
(480,000) pixels



(b) Image composed of 4,000
superpixels



(c) Superpixel image

9.1 Clustering and superpixels based segmentation

- Region segmentation using superpixels

- **SLIC Superpixel Algorithm** (*simple linear iterative clustering*)
- SLIC is a modification of the k -means algorithm
- SLIC observations typically use (but are not limited to) 5-dimensional vectors containing three color components and two spatial coordinates

9.1 Clustering and superpixels based segmentation

- Region segmentation using superpixels

- **SLIC Superpixel Algorithm** (*simple linear iterative clustering*)

1. Initialize the algorithm: Compute the initial superpixel cluster centers

$$\mathbf{m}_i = [r_i \ g_i \ b_i \ x_i \ y_i]^T, \quad i = 1, 2, \dots, n_{sp}$$

by sampling the image at regular grid steps, s . Move the cluster centers to the lowest gradient position in a 3×3 neighborhood. For each pixel location, p , in the image, set a label $L(p) = -1$ and a distance $d(p) = \infty$

2. Assign samples to cluster centers: For each cluster center $\mathbf{m}_i$, $i = 1, 2, \dots, n_{sp}$ compute the distance, $D_i(p)$ between $\mathbf{m}_i$ and *each* pixel p in a $2s \times 2s$ neighborhood about $\mathbf{m}_i$. Then, for each p and $i = 1, 2, \dots, n_{sp}$ if $D_i < d(p)$, let $d(p) = D_i$ and $L(p) = i$. n_{sp} denote the desired number of superpixels

9.1 Clustering and superpixels based segmentation

- Region segmentation using superpixels

- **SLIC Superpixel Algorithm** (*simple linear iterative clustering*)

3. Update the cluster centers: Let C_i denote the set of pixels in the image with label $L(p) = i$. Update m_i

$$m_i = \frac{1}{|C_i|} \sum_{z \in C_i} z \quad i = 1, 2, \dots, n_{sp}$$

where $|C_i|$ is the number of pixels in set C_i

4. Test for convergence: Compute the Euclidean norms of the differences between the mean vectors in the current and previous steps. Compute the residual error, E , as the sum of the n_{sp} norms. If $E < T$, where T a specified nonnegative threshold, go to Step 5. Else, go back to Step 2.

5. Post-process the superpixel regions: Replace all the superpixels in each region, C_i , by their average value, m_i .

9.1 Clustering and superpixels based segmentation

- Region segmentation using superpixels

- **SLIC Superpixel Algorithm** (*simple linear iterative clustering*)
- Specifying the distance measure: Let d_c and d_s denote the color and spatial Euclidean distances between two points in a cluster

$$d_c = \left[(r_j - r_i)^2 + (g_j - g_i)^2 + (b_j - b_i)^2 \right]^{1/2} \quad d_s = \left[(x_j - x_i)^2 + (y_j - y_i)^2 \right]^{1/2}$$

- Define D as the composite distance

$$D = \left[\left(\frac{d_c}{d_{cm}} \right)^2 + \left(\frac{d_s}{d_{sm}} \right)^2 \right]^{1/2}$$

where d_{cm} and d_{sm} are the maximum expected values of d_c and d_s

$$d_{sm} = s = \left[n_{tp} / n_{sp} \right]^{1/2} \quad \text{and} \quad d_{cm} = c$$

9.1 Clustering and superpixels based segmentation

- Region segmentation using superpixels

- **SLIC Superpixel Algorithm** (*simple linear iterative clustering*)

- The composite distance can be rewritten

$$\Rightarrow D = \left[\left(\frac{d_c}{c} \right)^2 + \left(\frac{d_s}{s} \right)^2 \right]^{1/2} = \left[d_c^2 + \left(\frac{d_s}{s} \right)^2 c^2 \right]^{1/2}$$

- Constant c can be used to weigh the relative importance between color similarity and spatial proximity.
 - c is large, spatial proximity is more important, and the resulting superpixels are more compact.
 - c is small, the resulting superpixels adhere more tightly to image boundaries, but have less regular size and shape.

9.2 Segmentation using graph cuts

- Expressing the pixels of the image as nodes of a graph, and then finding an optimum partition (*cut*) of the graph into groups of nodes.
- Optimality is based on criteria whose values are high for members within a group and low across members of different groups.
- Superior to the results achievable by any of the segmentation methods. The price of this potential benefit is added complexity in implementation, which generally translates into slower execution

9.2 Segmentation using graph cuts

- A graph, G , is a mathematical structure consisting of a set V of *nodes* and a set E of *edges* connecting those vertices $G = (V, E)$ where V is a set and $E \subseteq V \times V$ is a set of ordered pairs of elements from V
- *Undirected* graph: $(u, v) \in E$ implies that $(v, u) \in E$. Otherwise is *directed* graph. Image processing: *undirected* graph
- Edges are further characterized by a matrix, W , whose element $w(i, j) = w(j, i)$ is a weight associated with the edge that connects nodes i and j
- The weights are selected to be proportional to one or more similarity measures between all pairs of nodes

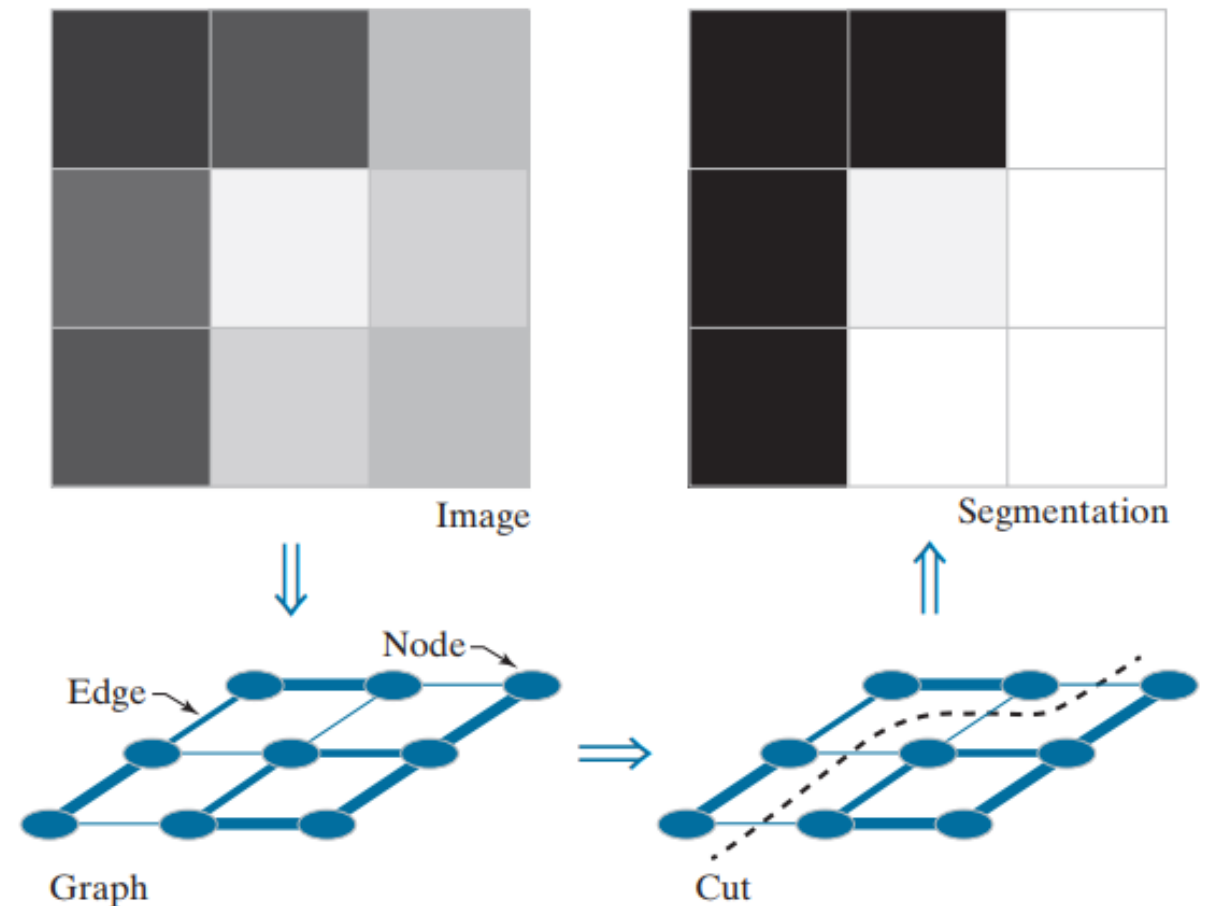
9.2 Segmentation using graph cuts

- A graph whose edges are associated with weights is called a *weighted graph*
- The weight, $w(i, j)$, of each edge is a function of the similarity between nodes i and j
- Seek to partition the nodes of the graph into disjoint subsets $V_1, V_2, \dots, V_K$ where, by some measure, the similarity among the nodes within a subset is high, and the similarity across the nodes of different subsets is low
- A *cut* of a graph is a partition of V into two subsets A and B such that

$$A \cup B = V \text{ and } A \cap B = \emptyset$$

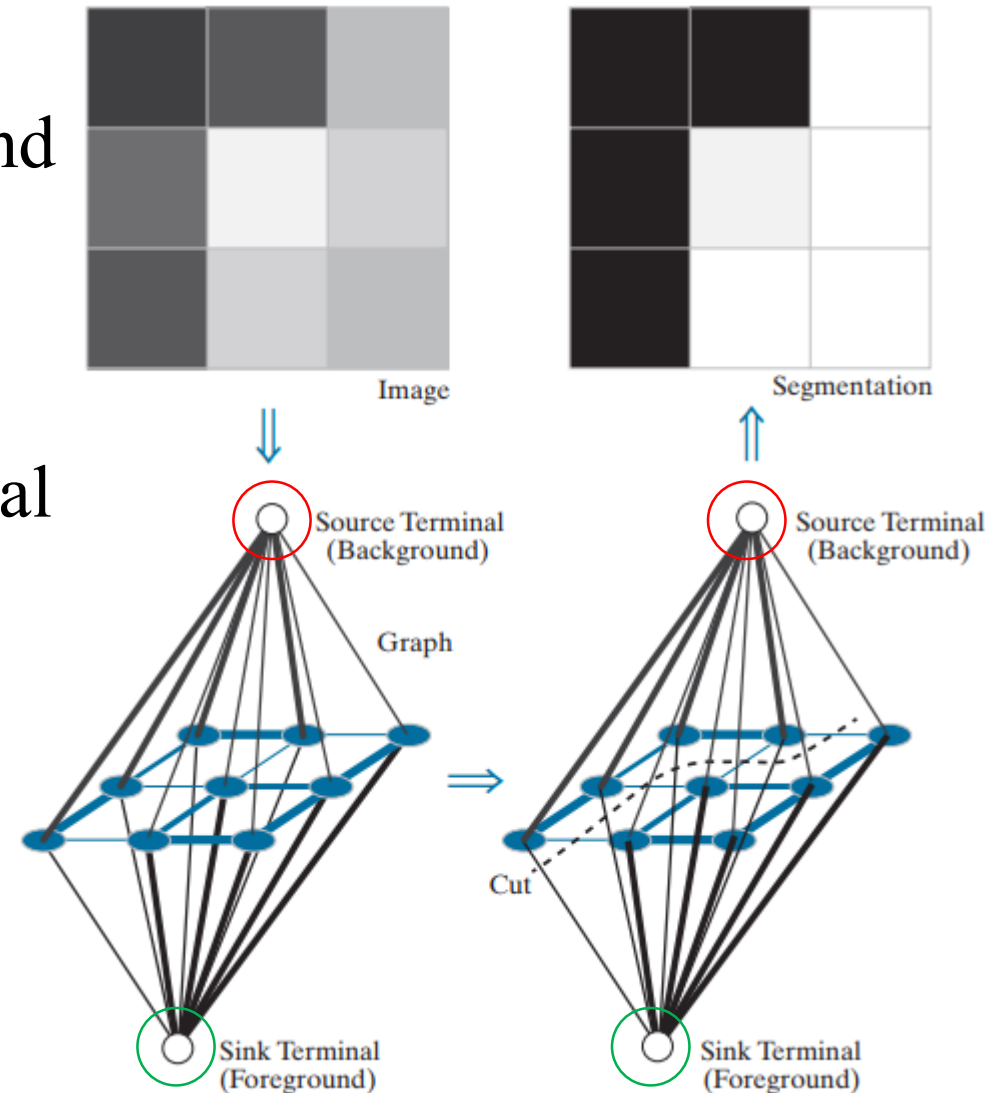
9.2 Segmentation using graph cuts

- Node = Pixel
- Edge = link between nodes



9.2 Segmentation using graph cuts

- Two additional nodes called the *source* and *sink terminal* nodes, respectively, each connected to all nodes in the graph via unidirectional links called *t-links*
- The thickness of each t-link is proportional to the value of the probability that the graph node to which it is connected is a foreground or background pixel



9.2 Segmentation using graph cuts - Minimum graph cuts

- This minimum cut is defined as the smallest *total* weight of the edges that, if removed, would disconnect the sink from the source

$$cut(A, B) = \sum_{u \in A, v \in B} w(u, v)$$

- The optimum partition of a graph is the one that minimizes this cut value
- Instead of the total weight value of the edges, a measure of “disassociation” that computes the cost as a fraction of the total edge connections to all nodes in the graph is used {*normalized cut (Ncut)*}

$$Ncut(A, B) = \frac{cut(A, B)}{assoc(A, V)} + \frac{cut(B, A)}{assoc(B, V)} \quad assoc(A, V) = \sum_{u \in A, z \in V} w(u, z)$$

9.2 Segmentation using graph cuts - Minimum graph cuts

- a measure for total *normalized association* within graph partitions

$$N_{assoc}(A, B) = \frac{assoc(A, A)}{assoc(A, V)} + \frac{assoc(B, B)}{assoc(B, V)}$$

where $assoc(A, A)$ and $assoc(B, B)$ are the total weights connecting the nodes within A and within B

- Then

$$N_{cut}(A, B) = 2 - N_{assoc}(A, B)$$

9.2 Segmentation using graph cuts - Computing minimal graph cuts

- The sum of the weights from node i to all other nodes in V : $d_i = \sum_j w(i, j)$
- *Normalized cut (Ncut)*

$$\begin{aligned} Ncut(A, B) &= \frac{cut(A, B)}{assoc(A, V)} + \frac{cut(A, B)}{assoc(B, V)} \\ &= \frac{\sum_{x_i > 0, x_j < 0} -w(i, j)x_i x_j}{\sum_{x_i > 0} d_i} + \frac{\sum_{x_i < 0, x_j > 0} -w(i, j)x_i x_j}{\sum_{x_i < 0} d_i} \end{aligned}$$

- Objective

$$\mathbf{x}_{\text{opt}} = \arg \min_{\mathbf{x}} Ncut(A, B)$$

9.2 Segmentation using graph cuts - Computing minimal graph cuts

- Solution

9.2 Segmentation using graph cuts – Graph cut segmentation algorithm

1. Given a set of features, specify a weighted graph, $G=(V,E)$ in which V contains the points in the feature space, and E contains the edges of the graph. Compute the edge weights and use them to construct matrices W and D . Let K denote the desired number of partitions of the graph.
2. Solve the eigenvalue system $(D - W)y = \lambda Dy$ to find the eigenvector with the second smallest eigenvalue
3. Use the eigenvector from Step 2 to bipartition the graph by finding the splitting point such that $Ncut(A, B)$ is minimized
4. If the number of cuts has not reached K , decide if the current partition should be subdivided by checking the stability of the cut
5. Recursively repartition the segmented parts if necessary

9.3 The use of motion in segmentation – Spatial Techniques

- **A basic approach**
- One of the simplest approaches for detecting changes between two image frames $f(x, y, t_i)$ and $f(x, y, t_j)$ taken at times t_i and t_j , respectively, is to compare the two images pixel by pixel
- A *difference image* of two images (of the same size) taken at times t_i and t_j

$$d_{ij}(x, y) = \begin{cases} 1 & \text{if } |f(x, y, t_i) - f(x, y, t_j)| > T \\ 0 & \text{otherwise} \end{cases}$$

9.3 The use of motion in segmentation – Spatial Techniques

- **Accumulative Differences**

- An *accumulative difference image* (ADI) is formed by comparing this reference image with every subsequent image in the sequence
- Let $R(x,y)$ denote the reference image and, k denote t_k so that $f(x, y, k) = f(x, y, t_k)$ We assume that $R(x,y)=f(x,y,1)$. For any $k>1$

$$A_k(x, y) = \begin{cases} A_{k-1}(x, y) + 1 & \text{if } |R(x, y) - f(x, y, k)| > T \\ A_{k-1}(x, y) & \text{otherwise} \end{cases}$$

$$P_k(x, y) = \begin{cases} P_{k-1}(x, y) + 1 & \text{if } |R(x, y) - f(x, y, k)| > T \\ P_{k-1}(x, y) & \text{otherwise} \end{cases}$$

$$N_k(x, y) = \begin{cases} N_{k-1}(x, y) + 1 & \text{if } |R(x, y) - f(x, y, k)| < -T \\ N_{k-1}(x, y) & \text{otherwise} \end{cases}$$

9.3 The use of motion in segmentation – Frequency Domain Techniques

- Consider a sequence $f(x, y, t)$, $t = 0, 1, \dots, K-1$, of K digital image frames of size $M \times N$ pixels, generated by a stationary camera
- Suppose that for frame one ($t = 0$) the object is at location (x', y') and the image plane is projected onto the x -axis; that is, the pixel intensities are summed (for each row) across the columns in the image
- Suppose that in frame two ($t = 0$) the object has moved to coordinates $(x'+1, y')$ that is, it has moved 1 pixel parallel to the x -axis
- If the object continues to move 1 pixel location per frame then, at any integer instant of time, t , the result will be

$$e^{j2\pi a_1(x'+t)\Delta t} = \cos[2\pi a_1(x' + t)\Delta t] + j \sin[2\pi a_1(x' + t)\Delta t]$$

9.3 The use of motion in segmentation – Frequency Domain Techniques

- Problem: determining motion via a Fourier transform formulation
- This procedure yields a complex sinusoid with frequency $a_1 \rightarrow$ If the object were moving pixels (in the x -direction) between frames, the sinusoid would have frequency $V_1 a_1 \rightarrow$ Thus a peak search in the Fourier spectrum would yield one peak with value $V_1 a_1$

9.3 The use of motion in segmentation – Frequency Domain Techniques

- For a sequence of k digital images of size $M \times N$ pixels, the sum of the weighted projections onto the x - and y -axis at any integer instant of time is

$$g_x(t, a_1) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y, t) e^{j2\pi a_1 x \Delta t} \quad t = 0, 1, \dots, K-1$$

$$g_y(t, a_2) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y, t) e^{j2\pi a_2 y \Delta t} \quad t = 0, 1, \dots, K-1$$

- 1D Fourier transforms

$$G_x(u_1, a_1) = \sum_{t=0}^{K-1} g_x(t, a_1) e^{-j2\pi u_1 t/K} \quad u_1 = 0, 1, \dots, K-1$$

$$G_y(u_2, a_2) = \sum_{t=0}^{K-1} g_y(t, a_2) e^{-j2\pi u_2 t/K} \quad u_2 = 0, 1, \dots, K-1$$

9.3 The use of motion in segmentation – Frequency Domain Techniques

- The frequency-velocity relationship $u_1 = a_1 V_1$ and $u_2 = a_2 V_2$
- The sign of the x -component of the velocity is obtained by computing

$$S_{1x} = \frac{d^2 \Re[g_x(t, a_1)]}{dt^2} \Big|_{t=n} \quad S_{2x} = \frac{d^2 \Im[g_x(t, a_1)]}{dt^2} \Big|_{t=n}$$

- The same sign at an arbitrary point in time, n , if the velocity component is positive and vice versa

9.4 Statistical Methods for Image Segmentation

- Given an image f_0 that is observed over the lattice Λ suppose that $\Lambda_1 \subseteq \Lambda$ and f_1 is a restriction of f_0 to only those pixels that belong to Λ_1

- Gaussian Statistics:** measures the amount of variability in the pixels that comprise f_1 along the (p, q) th direction.

$$T_{f_1}(p, q) = \sum_{(m, n) \in \Lambda_1} [f_1(m, n) - f_1(m + p, n + q)]^2 \text{ for } (p, q) \in \{(0, 1), (1, 0), \dots\}$$

- Fourier Statistics:** measures the amount of energy in frequency bin (c, f) that the pixels that comprise f possess.

$$F_{f_1}(\alpha, \beta) = \sum_{(m, n) \in \Lambda_1} [f_1(m, n)e^{-j(m\alpha + n\beta)}] [f_1(m, n)e^{j(m\alpha + n\beta)}] \text{ for } (\alpha, \beta) \in [-\pi, \pi]^2$$

- Covariance Statistics:** measures the correlation between the various components that comprise each pixel of j

$$K_{f_1} = \sum_{(m, n) \in \Lambda_1} ((f_1(m, n) - \mu_{f_1})^T ((f_1(m, n) - \mu_{f_1}) \text{ where } \mu_{f_1} = \sum_{(m, n) \in \Lambda_1} f_1(m, n)$$

- Label Statistics:** measures the amount of homogeneity in the pixels that comprise g_1 along the (p, q) th direction.

$$L_{g_1}(m, n) = \Psi[g_1(m, n), g_1(m + p, n + q)] \text{ where } \Psi(a, b) = \begin{cases} 1 & \text{if } a = b \\ -1 & \text{if } a \neq b \end{cases}$$

9.4 Statistical Methods for Image Segmentation

Some applications

- Vehicle Segmentation
- Aerial Image Segmentation
- Segmentation for Image Compression

9.5 Video Segmentation

- Video segmentation is an integral part of many video analysis and coding problems, including:
 - (i) video summarization, indexing and retrieval
 - (ii) advanced video coding
 - (iii) video authoring and editing
 - (iv) improved motion (optical flow) estimation
 - (v) 3D motion and structure estimation with multiple moving objects
- Video segmentation refers to partitioning video into spatial, temporal or spatio-temporal regions that are homogeneous in some feature space

9.5 Video Segmentation

- Effective video segmentation requires proper feature selection and an appropriate distance measure
- No guarantee that any of the resulting automatic segmentations will be semantically meaningful, since a semantically meaningful region may have multiple colors, multiple textures and/or multiple motion
- Factors that affect the choice of a specific segmentation method include
 - Real-time performance
 - Precision of segmentation
 - Scene complexity

9.5 Video Segmentation – Scene Change Detection

- It is one-dimensional, along the temporal dimension, segmentation problem
- Shot boundary detection methods locate temporal discontinuities, i.e., frames across which large differences are observed in some feature space, usually a combination of color and motion
- The simplest approach for detecting temporal discontinuities is to quantify frame differences in the pixel intensity domain → If a pre-determined number of pixels exhibit differences larger than a threshold value, then a cut can be declared

9.5 Video Segmentation – Scene Change Detection

Several ways to improve robustness

- Divide each frame into rectangular blocks, compute statistics of each block such as the mean and variance independently, and then check the count of blocks with changing statistics against a set threshold.
- Applying low-pass filtering to each frame prior to computing frame differences or block statistics.
- consider frame histogram differences instead of pixelwise or blockwise frame intensity differences.

9.5 Video Segmentation – Spatio-Temporal Change Detection

- Change detection methods segment each frame into two regions, namely changed and unchanged regions in the case of a static camera or global and local motion regions in the case of a moving camera
 - unchanged regions correspond to the background (null hypothesis) and changed regions to the foreground object(s) or uncovered (occlusion) areas
- Discussion: What happens to moving camera?
- Evaluate change detection methods
 - what features and background model are used
 - what distance metrics are used
 - what kind of threshold and background model adaptation rules are used

9.5 Video Segmentation – Spatio-Temporal Change Detection

- **Spatial Change Detection Using Two Frames**
- The simplest method to detect changes between two registered frames would be to analyze the frame difference (FD) image

$$FD_{k,r}(\mathbf{p}) = s(\mathbf{p}, k) - s(\mathbf{p}, r)$$

where $\mathbf{p}=(m,n)$ is pixel location and $s(\mathbf{p},k)$ is the intensity at pixel $\mathbf{p}$ in frame k

- It shows the pixel-to-pixel difference between the current image k and the reference image r

9.5 Video Segmentation – Spatio-Temporal Change Detection

- **Spatial Change Detection Using Two Frames**
- In order to distinguish the nonzero differences that are due to noise from those that are due to local motion, segmentation can be achieved by thresholding the FD as

$$r_{k,r}(\mathbf{p}) = \begin{cases} 1 & \text{if } |\text{FD}_{k,r}(\mathbf{p})| > T \\ 0 & \text{otherwise} \end{cases}$$

where T is an appropriate threshold. Here, $r_{k,r}(\mathbf{p})$ is called a segmentation label field, which is equal to “1” for changed regions and “0” otherwise.

9.5 Video Segmentation – Spatio-Temporal Change Detection

- **Spatial Change Detection Using Two Frames**
- An improved change detection algorithm
 - i. Construct a Gaussian pyramid where each frame is represented in multiple resolutions. Start processing at the lowest resolution level
 - ii. For each pixel at the present resolution level, compute the normalized frame difference given by

$$\text{FDN}_{k,r}(\mathbf{p}) = \frac{\sum_{\mathbf{p} \in \mathcal{N}} |s(\mathbf{p}, k) - s(\mathbf{p}, r)| |\nabla s(\mathbf{p}, r)|}{\sum_{\mathbf{p} \in \mathcal{N}} |\nabla s(\mathbf{p}, r)|^2 + c}$$

- iii. Repeat step (ii) for all resolution levels

9.5 Video Segmentation – Spatio-Temporal Change Detection

- **Temporal Integration**
- An important consideration is to add memory to the motion process in order to ensure both spatial and temporal continuity of the changed regions at each frame
 - temporal filtering (integration) of the intensity values across multiple frames before thresholding
 - post-processing of labels after thresholding
- A variation of the successive frame difference and normalized frame difference is the frame difference with memory
$$\text{FDM}_k(\mathbf{p}) = s(\mathbf{p}, k) - \bar{s}(\mathbf{p}, k)$$
- where $\bar{s}(\mathbf{p}, k) = (1 - \alpha)s(\mathbf{p}, k) + \alpha\bar{s}(\mathbf{p}, k - 1), \quad k = 1, \dots; 0 < \alpha < 1$

9.5 Video Segmentation – Motion Segmentation

- Motion segmentation (also known as optical flow segmentation) methods label pixels (or optical flow vectors) at each frame that are associated with independently moving part of a scene.
- Motion segmentation is closely related to two other problems, motion (change) detection and motion estimation.
- An important distinction between change detection and motion segmentation
 - change detection can be achieved without motion estimation if the scene is recorded with a static camera. For moving camera and general motion segmentation, it requires some sort of global and/or local motion estimation

9.5 Video Segmentation – Motion Segmentation

- **Segmentation Using Two Frames**

- i. Compute the dominant 2D translation vector (d_x, d_y) over the whole frame as the solution of

$$\begin{bmatrix} \sum I_x^2 & \sum I_x I_y \\ \sum I_x I_y & \sum I_y^2 \end{bmatrix} \begin{bmatrix} d_x \\ d_y \end{bmatrix} = \begin{bmatrix} -\sum I_x I_t \\ -\sum I_y I_t \end{bmatrix}$$

where I_x, I_y, I_t denote partials of image intensity with respect to x, y , and t .

9.5 Video Segmentation – Motion Segmentation

- **Segmentation Using Two Frames**

- ii. Label all pixels that correspond to the estimated dominant motion
 - a) Register the two images using the estimated dominant motion model. The dominant object appears stationary between the registered images, while other parts of the image are not.
 - b) Then, the problem reduces to labeling stationary regions between the registered images, which can be solved by the multiresolution change detection algorithm
 - c) In addition to the normalized frame difference, define a motion reliability measure as the reciprocal of the condition number of the coefficient matrix

$$R(x, k) = \frac{\lambda_{min}}{\lambda_{max}}$$

the smallest and largest
eigenvalue of the
coefficient matrix

9.5 Video Segmentation – Motion Segmentation

- iii. Estimate the parameters of a higher order motion model (affine, perspective, or quadratic) over the new region of analysis. Iterate over steps (ii) and (iii) until a satisfactory segmentation is attained.

9.5 Video Segmentation – Multiple Motion Segmentation

- **Clustering in the Motion Parameter Space**
 - K-means Method
 - Hough Transform Methods

9.6 2D and 3D Motion Tracking in Digital Video - Rigid Object Tracking

Two-Dimensional Rigid Object Tracking

- Remind from CPV301: Methods for 2D rigid object tracking can be classified in different categories
 - Region-based methods
 - Contour-based methods
 - Feature-based methods
 - Template-based methods

9.6 2D and 3D Motion Tracking in Digital Video - Rigid Object Tracking

Two-Dimensional Rigid Object Tracking

Bayesian Object Tracking

- Evolution of the object state over time

$$\mathbf{x}_k = f_k(\mathbf{x}_{k-1}, \mathbf{w}_{k-1})$$

- Measurement model, linking noisy measurements to the state vector

$$\mathbf{z}_k = h_k(\mathbf{x}_k, \mathbf{v}_k)$$

9.6 2D and 3D Motion Tracking in Digital Video - Rigid Object Tracking

Two-Dimensional Rigid Object Tracking

Bayesian Object Tracking

- Posterior at time step $k - 1$ is propagated forward in time, using the system model

$$p(\mathbf{x}_k | \mathbf{z}_{k-1}) = \int p(\mathbf{x}_k | \mathbf{x}_{k-1}) p(\mathbf{x}_{k-1} | \mathbf{z}_{k-1}) d\mathbf{x}_{k-1}$$

- Desired posterior probability density function can be obtained by using Bayes theorem

$$p(\mathbf{x}_k | \mathbf{z}_k) = \frac{p(\mathbf{z}_k | \mathbf{x}_k) p(\mathbf{x}_k | \mathbf{z}_{k-1})}{p(\mathbf{z}_k | \mathbf{z}_{k-1})}$$

where $p(\mathbf{z}_k | \mathbf{z}_{k-1}) = \int p(\mathbf{z}_k | \mathbf{x}_k) p(\mathbf{x}_k | \mathbf{z}_{k-1}) d\mathbf{x}_k$

$$\hat{\mathbf{x}}_k = \arg \max_{\mathbf{x}_k} p(\mathbf{x}_k | \mathbf{z}_k)$$

9.6 2D and 3D Motion Tracking in Digital Video - Rigid Object Tracking

- **Three-Dimensional Rigid Object Tracking**
- Three-dimensional (3D) rigid object tracking involves estimating the position and orientation of a rigid object in 3D space from video sequences
- This process requires determining the object's position (center of mass) and orientation relative to a world coordinate system, resulting in six parameters to estimate
- A key application is 3D head tracking (head pose estimation), used in face recognition, facial expression analysis, and human-computer interaction. While the human head isn't rigid, treating it as such is often sufficient

9.6 2D and 3D Motion Tracking in Digital Video - Rigid Object Tracking

- **Three-Dimensional Rigid Object Tracking**
- A typical model-based 3D rigid object tracking algorithm
 - A 3D geometric model of the object is initialized and enriched with texture information.
 - Then, for each video frame the model parameter vector that best describes the object in the new frame is evaluated.
- Model parameters derived in previous frame are used to provide a rough estimate of the parameters in the current frame, → maximize the similarity between the projection of the model on the image plane and the underlying image content.

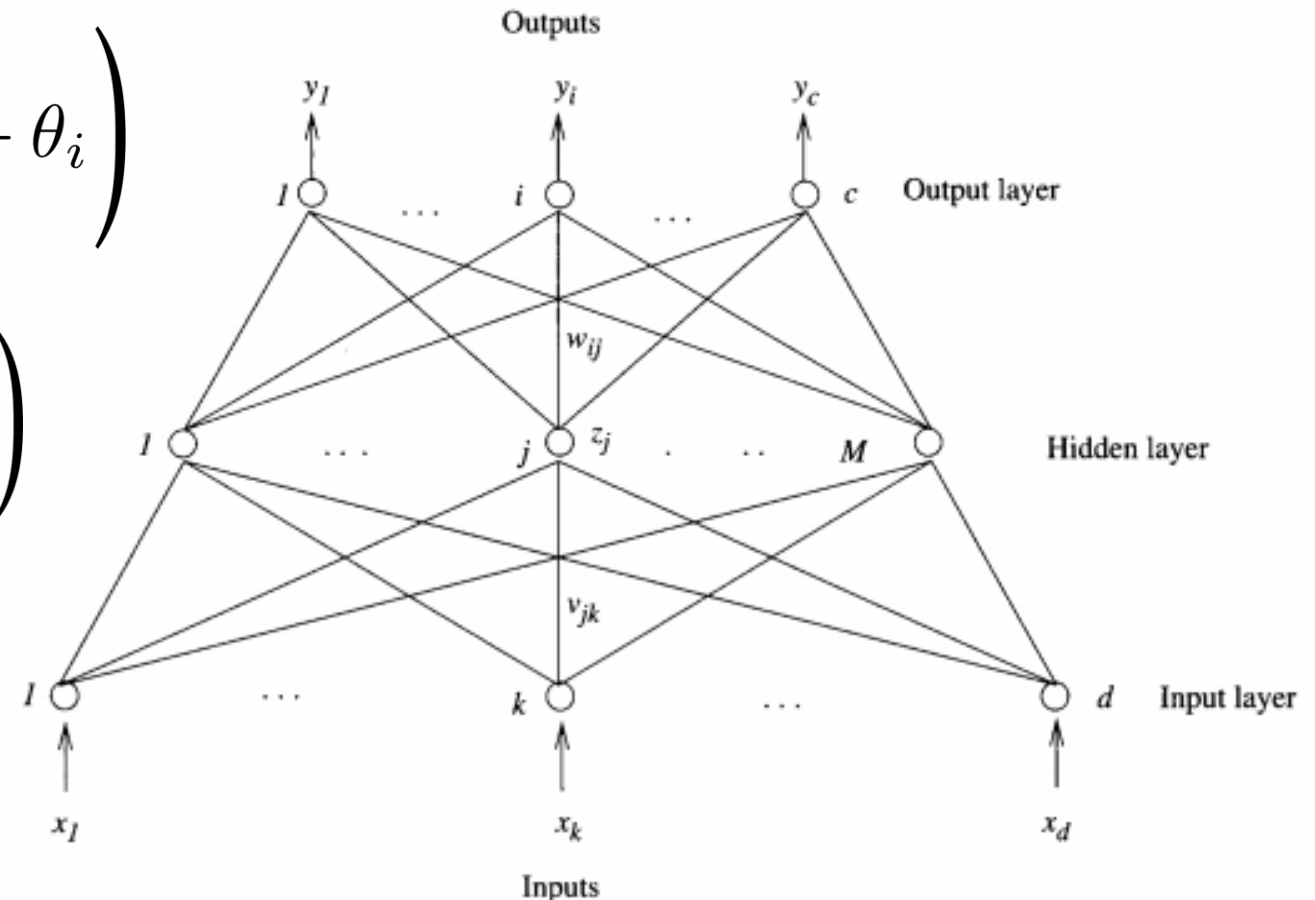
9.7 Adaptive and Neural Methods for Image Segmentation

- Topology of a feedforward ANN with a single hidden layer

$$y_i(\mathbf{x}) = f \left(\sum_{j=1}^M w_{ij} z_j(\mathbf{x}) + \theta_i \right)$$

$$z_j(\mathbf{x}) = g \left(\sum_{k=1}^d v_{jk} x_k + \theta_j \right)$$

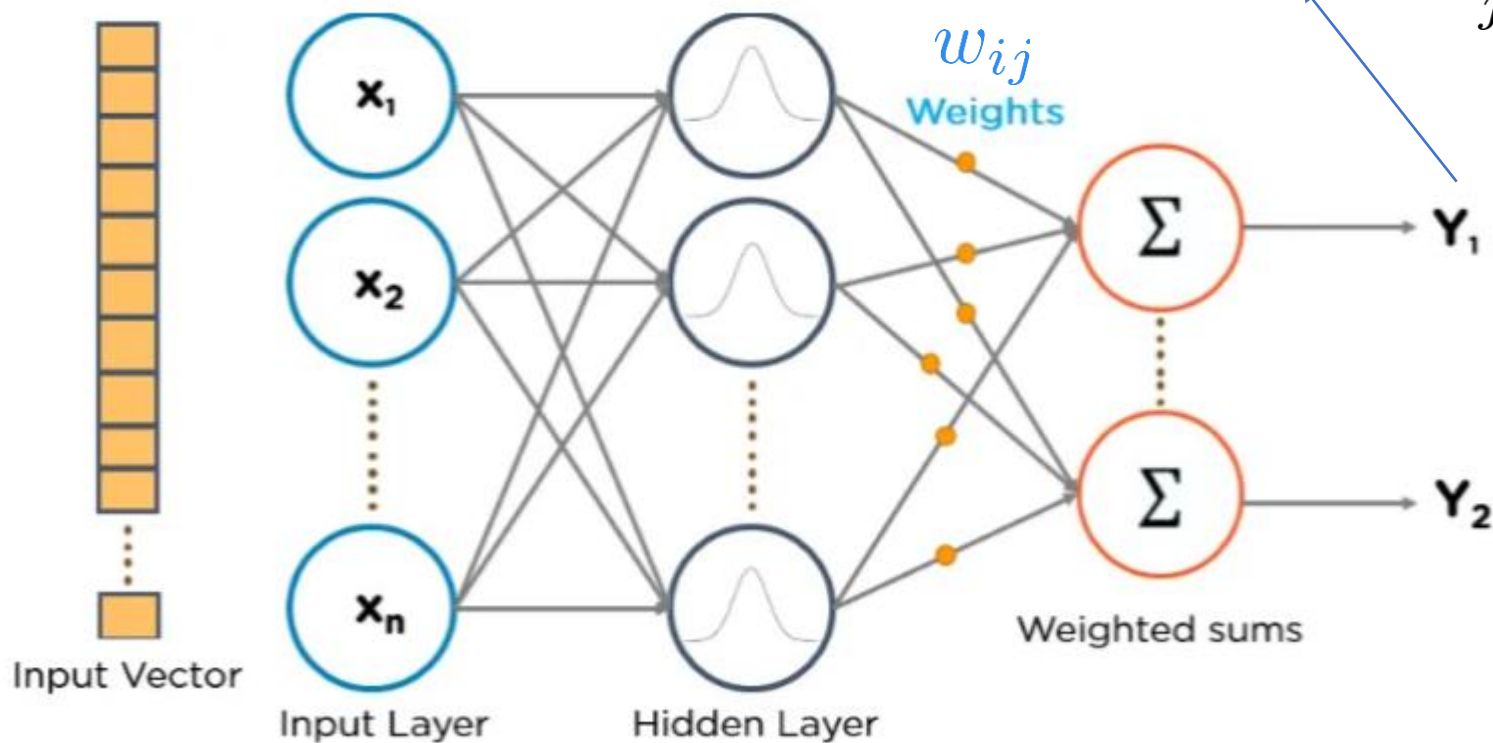
$$\mathbf{x} = [x_1, x_2, \dots, x_d]$$



9.7 Adaptive and Neural Methods for Image Segmentation

- Radial basis function network (RBFN)

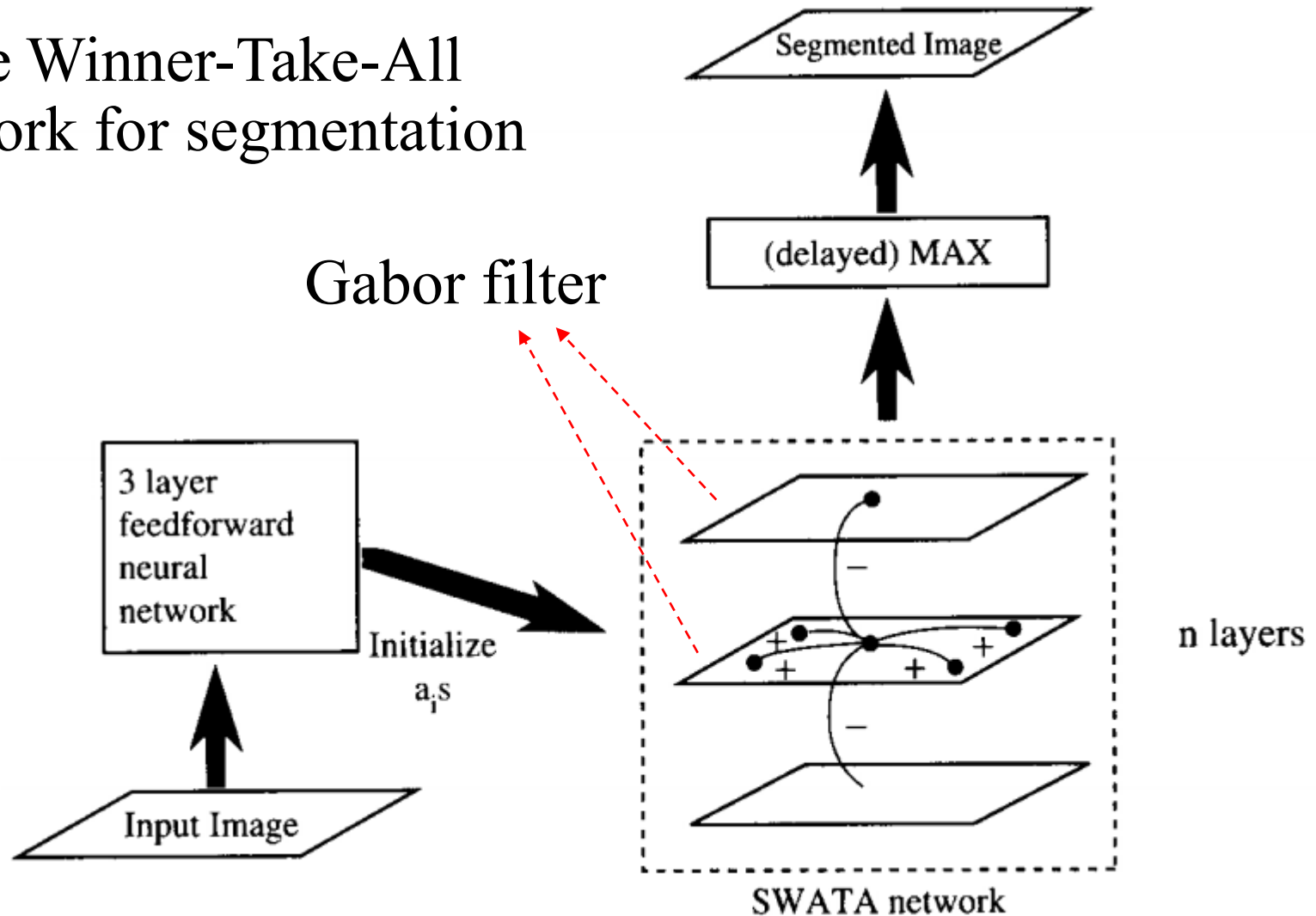
$$y_i(\mathbf{x}) = \sum_{j=1}^M w_{ij} \phi(\|\mathbf{x} - \mu_i\|)$$



$$\phi(\mathbf{x}) = e^{-\frac{\|\mathbf{x} - \mu_c\|^2}{\sigma^2}}$$

9.7 Adaptive and Neural Methods for Image Segmentation

- The Smoothing, Adaptive Winner-Take-All Network (SAWTA) network for segmentation of textured images

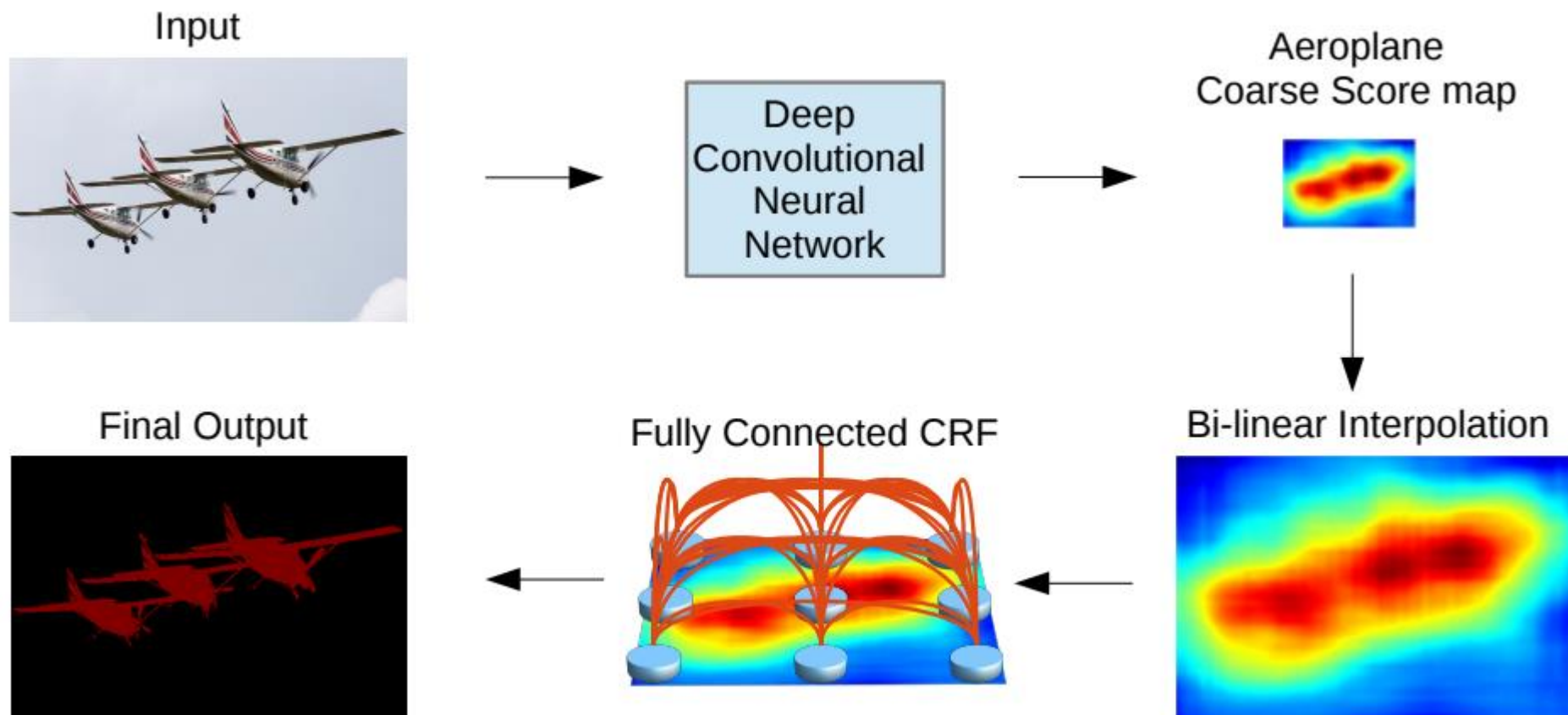


9.8 Machine learning/deep learning based segmentation

- Semantic segmentation
 - Full supervised methods
 - Semi-supervised methods
 - Weakly supervised methods
 - Unsupervised methods
- Instance segmentation
 - Multi-stage instance segmentation
 - Single-stage instance segmentation
 - Semi/weakly supervised instance segmentation

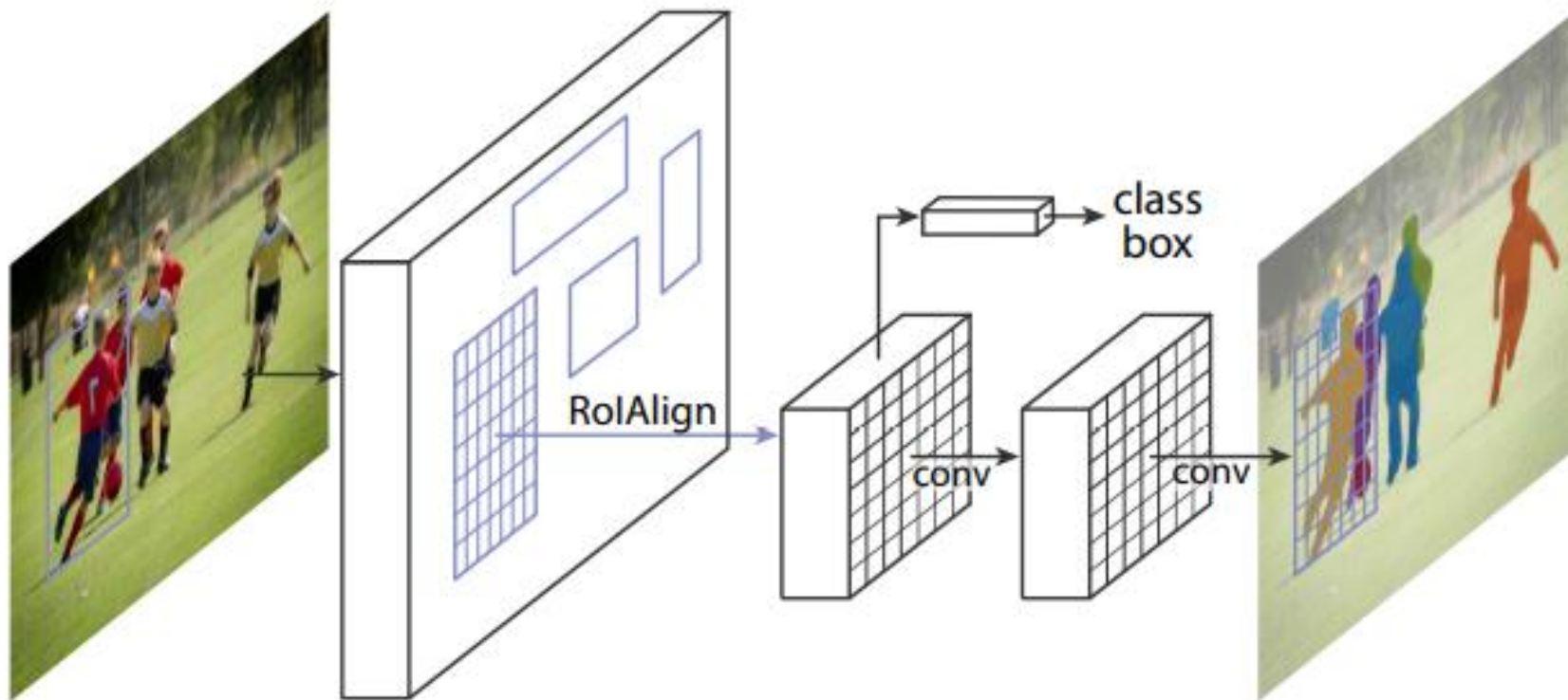
9.8 Machine learning/deep learning based segmentation

- DCNN: VGG16/ResNet50/...



9.8 Machine learning/deep learning based segmentation

- Mask RCNN



9.8 Machine learning/deep learning based segmentation

- Discussion: Deep learning based methods are probably better than traditional methods?

Summary

- Clustering and superpixels based segmentation
- Segmentation using graph cuts
- The use of motion in segmentation
- Statistical Methods for Image Segmentation
- Video Segmentation
- 2D and 3D Motion Tracking in Digital Video
- Adaptive and Neural Methods for Image Segmentation
- Machine learning/deep learning based segmentation